

Question-1

$$f(x) = (\sqrt{x})^{\log_5(x)-1}$$

(i) Find the domain of $f(x)$

as per the laws i.e.,

if $\sqrt[n]{x}$ then $x \geq 0$

if $\log_b(x)$, $x > 0$, $b > 0$, $b \neq 1$

Using above laws, our equation is divided into 2 parts, i.e. -

- $\sqrt{x}$, which derives $x \geq 0$, domain as $[0, \infty)$

- $\log_5(x)$, which derives $x > 0$, domain as $(0, \infty)$

So domain of full function, i.e. $f(x)$ is $(0, \infty)$ or $0 < x < \infty$ or $x > 0$

(ii) Solve equation $f(x) = 5$

$$\therefore f(x) = (\sqrt{x})^{\log_5(x)-1}$$

$$\text{So, } (\sqrt{x})^{\log_5(x)-1} = 5$$

Substituting, $\log_5$ both side

$$\log_5 (\sqrt{x})^{\log_5(x)-1} = \log_5 5$$

$$\therefore, \left. \begin{array}{l} \log_5 (x^n) = n \log_5(x) \\ \& \log_5(1) = 1 \end{array} \right\} \text{log properties}$$

So, using above properties

$$\begin{aligned} (\log_5(x)-1) (\log_5(x)^{\frac{1}{2}}) &= 1 \\ \frac{1}{2} \log_5 x \cdot \log_5 x - \frac{1}{2} \log_5 x &= 1 \end{aligned}$$

let $a = \log_5 x$
 so, $\frac{a}{2} \cdot a - \frac{a}{2} = 1$

$$a^2 - a - 2 = 0$$

$$a^2 + a - 2a - 2 = 0$$

$$(a+1)(a-2) = 0 \Rightarrow a = 2 \text{ or } a = -1$$

if $a = 2$, then $\log_5 x = 2$

$$5^2 = x$$

$$x = 25$$

, using $y = \log_5(x)$
 $x = 5^y$

if $a = -1$, then $\log_5 x = -1$

$$5^{-1} = x$$

$$x = \frac{1}{5}$$

So, for $f(x) = 5$, x is 25 or $\frac{1}{5}$

Question-2

$$f(x) = \frac{x}{1+x|x|}$$

(i) Find domain of $f(x)$

For these kind of func., we need to ensure denominator is not 0 bec. division by zero is undefined.

so, $1+x|x| \neq 0$

if $x \geq 0$,

$$1+x(x) \geq (1+x^2), \text{ domain} \Rightarrow [0, \infty)$$

if $x < 0$,

$$1+x(-x) \geq (1-x^2), \text{ domain} \Rightarrow (-\infty, 0) \text{ but } x \neq -1$$

so combining both domain for $f(x)$

$(-\infty, -1) \cup (-1, \infty)$ or All real nos. except $x = -1$.

$$x \in \mathbb{R}, x \neq -1$$

(ii) Find all x - & y -intercepts.

for x -intercept, $f(x)=0$

$$\frac{x}{1+x|x|} = 0$$

$$x = 0$$

for y -intercept, $x=0$

$$f(0) = \frac{0}{1+0|0|} = 0$$

So, x & y intercepts are 0 & 0 respectively.

(iii) Write func. as piecewise-defined func.

$$f(x) = \begin{cases} x/(1+x^2) & , x > 0 \\ 0 & , x = 0 \\ x/(1-x^2) & , -1 < x < 0 \\ \text{not defined} & , x = -1 \\ x/(1-x^2) & , x < -1 \end{cases}$$

(iv) For stationary pts., we need to find 1st derivative for $x \geq 0$

here, $x=1$, if $y'=0$

$$y = \frac{x}{1+x^2}$$

$$y' = \frac{(1+x^2) - x(2x)}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

for $x < 0$

$$y = \frac{x}{1-x^2}$$

here, $x=-1$, if $y'=0$
as undefined

$$y' = \frac{(1-x^2) - x(-2x)}{(1-x^2)^2} = \frac{1+x^2}{(1-x^2)^2}$$

So from above derivatives, x can be derived as ± 1

as, x can't be -1 . So stationary point is $(1, 0.5)$, which is only present.

$$f(1) = \frac{1}{1+1|1|} = 0.5$$

let's use 2nd derivative test to classify this point for $x \geq 0$

$$y' = \frac{1-x^2}{(1+x^2)^2}$$

$$y'' = \frac{-2x(1+x^2)^2 - (1-x^2)2(1+x^2)2x}{((1+x^2)^2)^2}$$

$$y'' = \frac{-2x(1+x^2)(1+x^2-2x^2)}{(1+x^2)^3(1+x^2)}$$

$$y'' = \frac{2x^3-6x}{(1+x^2)^3}$$

for $x < 0$

$$y' = \frac{1+x^2}{(1-x^2)^2}$$

$$y'' = \frac{2x(1-x^2)^2 - (1+x^2)2(1-x^2)(-2x)}{(1-x^2)^4}$$

$$y'' = \frac{2x(1-x^2)(1-x^2+2x^2+2)}{(1-x^2)^3(1-x^2)}$$

$$y'' = \frac{2x^3+6x}{(1-x^2)^3}$$

$f''(1)$, when $x \geq 0$

$$= \frac{2-6}{(1+1)^3} = -4 < 0$$

$\Rightarrow (1, 0.5)$ is local max. point.

(v) We use sign table of derivative to see when derivative is +ve or -ve which means when func. is increasing or decreasing

$x \geq 0$			$x < 0$		
0	1	∞	$-\infty$	-1	0
$1-x^2$	+	-	$1+x^2$	+	+
$(1+x^2)^2$	+	+	$(1-x^2)^2$	+	+
y'	+	-	y'	+	+

from above sign table for 1st derivative, we derive
Function is increasing on $(-\infty, 1)$ & it is decreasing
on $(1, \infty)$. $\rightarrow$ except (-1) as not defined

So intervals can be defined as:

increasing for $-\infty < x < 1$, $x \neq -1$
undefined for $x = -1$
decreasing for $1 < x < \infty$

(vi) As 2nd derivative is already calculated, i.e
for $x \geq 0$,

$$y'' = \frac{2x^3 - 6x}{(1+x^2)^3} \Rightarrow$$

here, x is derived as
 $2x(x^2 - 3)$
so, $x = 0$ or $\sqrt{3}$

for $x < 0$,

$$y'' = \frac{2x^3 + 6x}{(1-x^2)^3} \Rightarrow$$

here, x is derived as
 $2x(x^2 + 3)$
so, $x = -\sqrt{3}$

Using sign table for deriving func. is concave or convex.

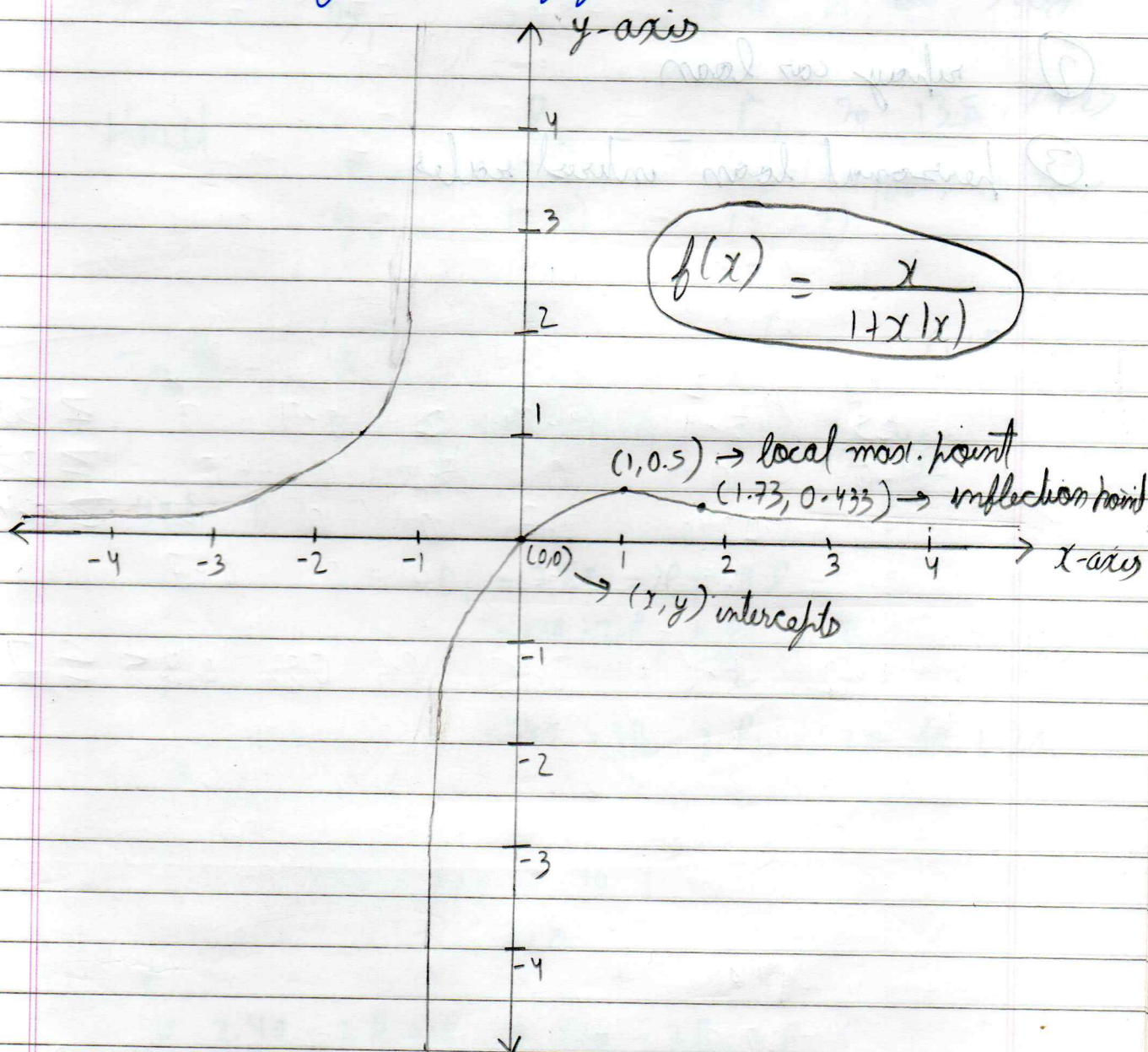
$x \geq 0$				$x < 0$				
0	$\sqrt{3}$	∞		$-\infty$	$-\sqrt{3}$	-1	0	
$2x^3 - 6x$	-	+		$2x^3 + 6x$	-	-	-	
$(1+x^2)^3$	+	+		$(1-x^2)^3$	-	-	+	
y''	0	-	0	y''	+	+	0	-

from above sign table for 2nd derivative, we derive

Convex or concave up $(-\sqrt{3}, \sqrt{3}), (-\infty, -1)$
Concave or concave down $(0, \sqrt{3}), (-1, 0)$
where $x \neq -1$

Convex in range, $\{ -\infty < x < -1 \} \cup \{ \sqrt{3} < x < \infty \}$
Concave in range, $\{ -1 < x < \sqrt{3} \}$
Undefined for, $x = -1$

(vii) Sketching graph of function



Inflection point, we need 2 conditions

① $y'' = 0$

② y'' should change ^{sign} value at a point

$x \geq 0, x = \sqrt{3}$

$$y' = \frac{\sqrt{3}}{1+\sqrt{3}|\sqrt{3}|} = \frac{\sqrt{3}}{4} = 0.433$$

$x < 0$

inflection point is $(\sqrt{3}, 0.433) \approx (1.732, 0.433)$